Data Management Technologies
Unit 10 Consolidation task

Activity 1.: Administering a Database

Task 1: Database Initialization

Create a simple database in MySQL or MongoDB.
Define appropriate users and assign roles (e.g., admin, read-only, data entry).
You should supply screenshots showing database creation and role assignments.

Response

Overview of database created

This is the database for a small company with details of 10 staff, including names, employment dates and pay details.

	staff_id	first_name	last_name	salary	employment_date	department
Edit Copy Delete	1	Alice	Bartholomew	45000	2023-06-05 09:00:00	Engineering
Edit Copy Delete	2	Bobby	Baker	62000	2025-06-02 09:00:00	Marketing
Edit Copy Delete	3	Carol	Barber	58000	2024-12-13 22:30:24	HR
Edit Copy Delete	4	Eamon	Arnold	30000	2024-11-11 08:48:05	Administration
Edit Copy Delete	5	Lotus	Olso	45000	2025-12-08 06:50:18	Finance
Edit Copy Delete	6	Iris	Banbury	60000	2025-10-06 09:00:00	HR
Edit Copy Delete	7	Daffodil	Telford	45000	2024-04-08 04:50:18	Administration
Edit Copy Delete	8	Ade	Elegushi	34000	2025-02-10 09:00:08	Marketing
Edit Copy Delete	9	Johnny	Elland	23000	2026-01-05 22:50:18	Finance
Edit Copy Delete	10	Maria	Alderman	60000	2025-08-11 22:51:03	Executive

Check all *With selected:* Edit Copy Delete Export

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Image 1: Main table 'staff"

Admin users have full access (including access to DDL and DML). Can create and remove tables in SQL, using CREATE, DROP and DELETE. This is granted to administrative staff.

Read-only users have access to the SELECT keyword only, so that they can view, but not edit the data. This is granted to managers, executive staff and auditors, who require the information, but have no need to change it.

Data entry users have access to SELECT, INSERT and UPDATE keywords, so can edit the data. This includes HR staff, clerks, and receptionists who require moderate permissions to change the data.

The SQL syntax for role assignments is as below:

```sql
-- Create users
CREATE USER IF NOT EXISTS 'Daffodil_Telford_Admin'@'localhost'     IDENTIFIED BY 'AdminPass123!';
CREATE USER IF NOT EXISTS 'Maria_Alderman_Executive'@'localhost'  IDENTIFIED BY 'ReadOnly123!';
CREATE USER IF NOT EXISTS 'Carol_Barber_HR'@'localhost' IDENTIFIED BY 'DataEntry123!';

-- Assign roles/privileges

-- Admin: full control
GRANT ALL PRIVILEGES ON unit_10-consolidation_db.* TO 'admin_user'@'localhost';

-- Read-only: SELECT only
GRANT SELECT ON unit_10-consolidation_db.*  TO 'readonly_user'@'localhost';

-- Data entry: INSERT and UPDATE only (no DELETE, no schema changes)
GRANT SELECT, INSERT, UPDATE ON unit_10-consolidation_db.* TO 'dataentry_user'@'localhost';

FLUSH PRIVILEGES;

-- Print permission information

SHOW GRANTS FOR 'admin_user'@'localhost';
SHOW GRANTS FOR 'readonly_user'@'localhost';
SHOW GRANTS FOR 'dataentry_user'@'localhost';
```

Image 2: Role Assignments

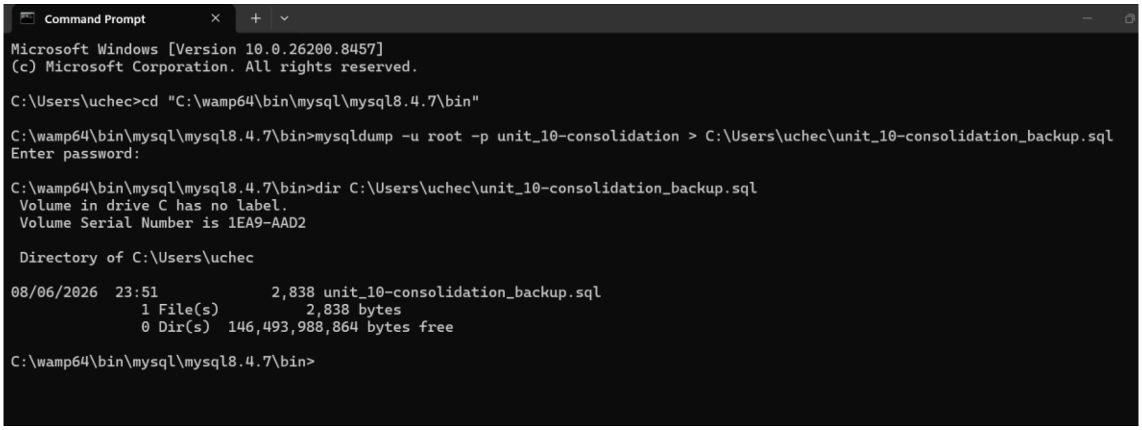
Task 2: Backups and Restorations:

Use tools (e.g., mysqldump for MySQL or mongodump for MongoDB) to back up a sample database. Simulate data loss by deleting a table or collection and restore the database from the backup. Document: backup and restoration commands, outputs, and a reflection on the process.

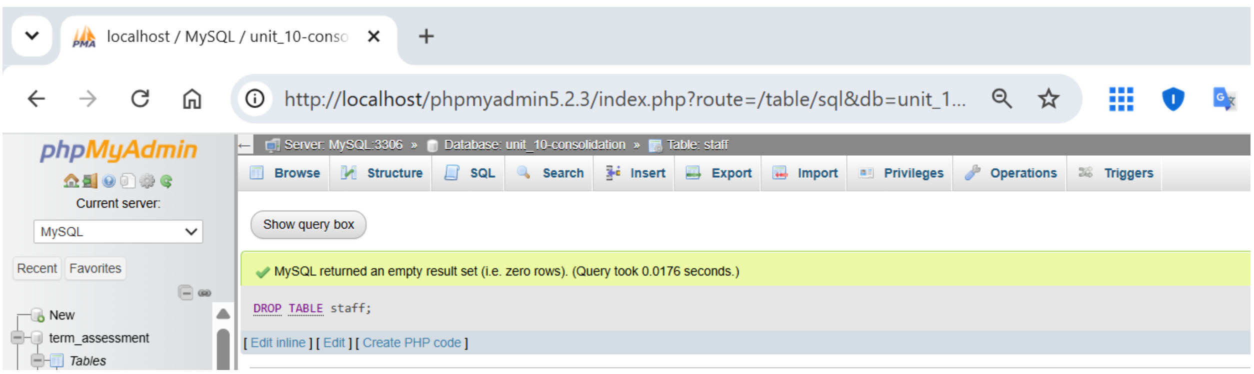
Response:

Simulation of data loss: The steps and screenshots are recorded below:

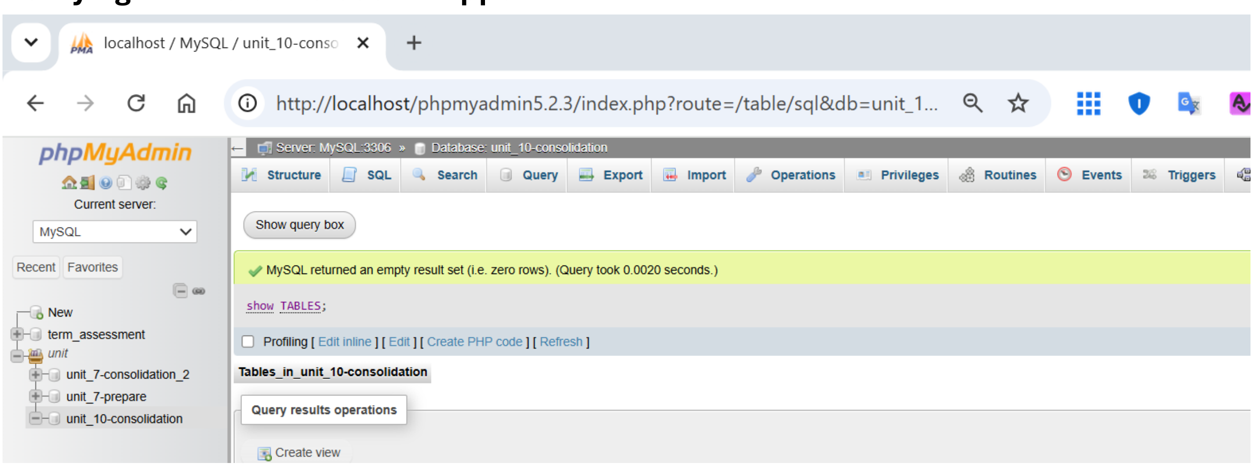
Back up on command prompt:



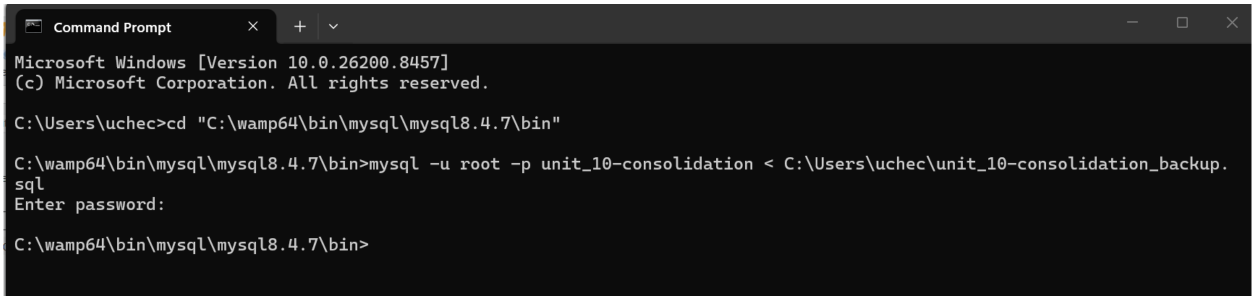
Testing backup by dropping the table staff in phpMYAdmin:



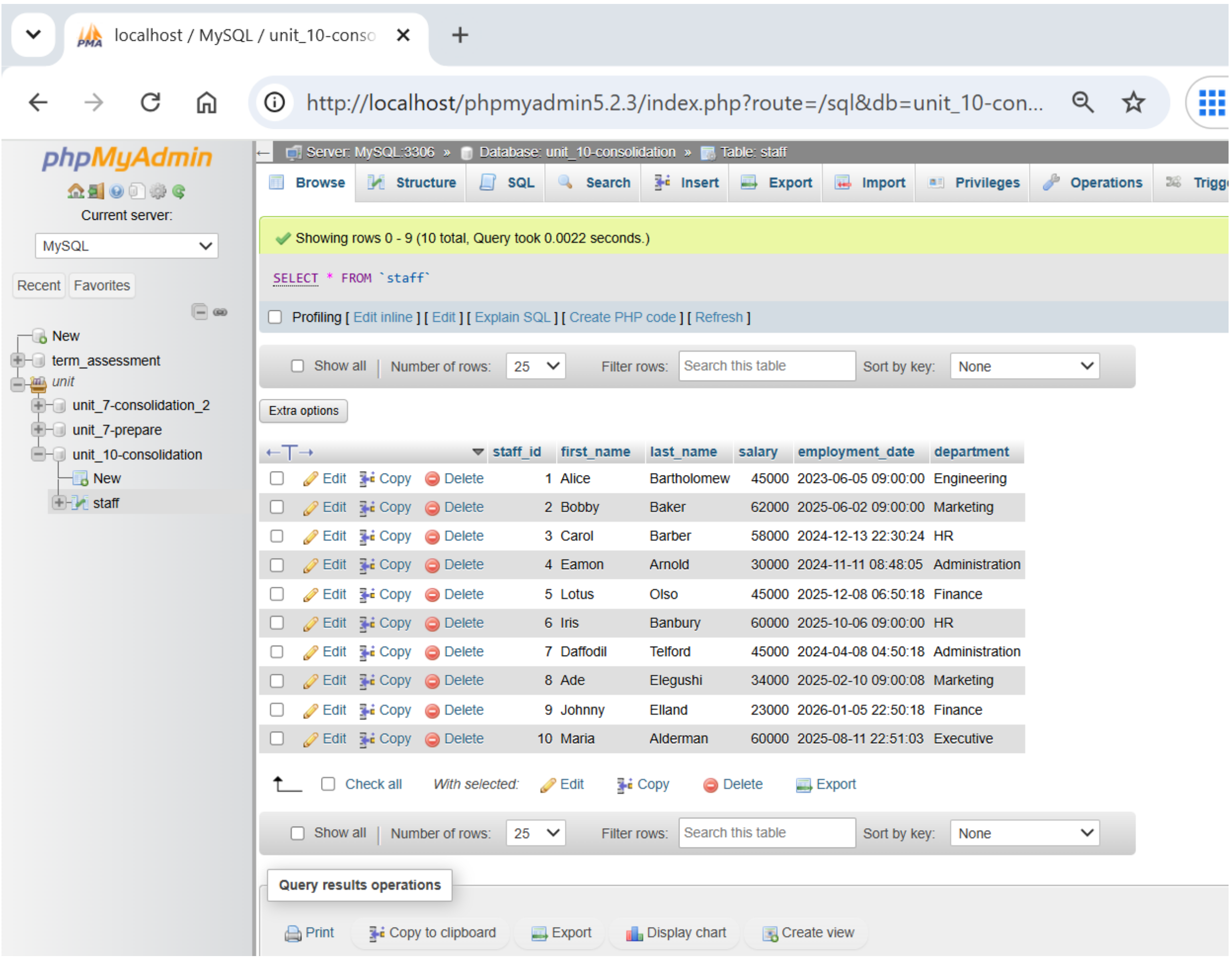
Verifying the table has been dropped:



Restoring data via command prompt



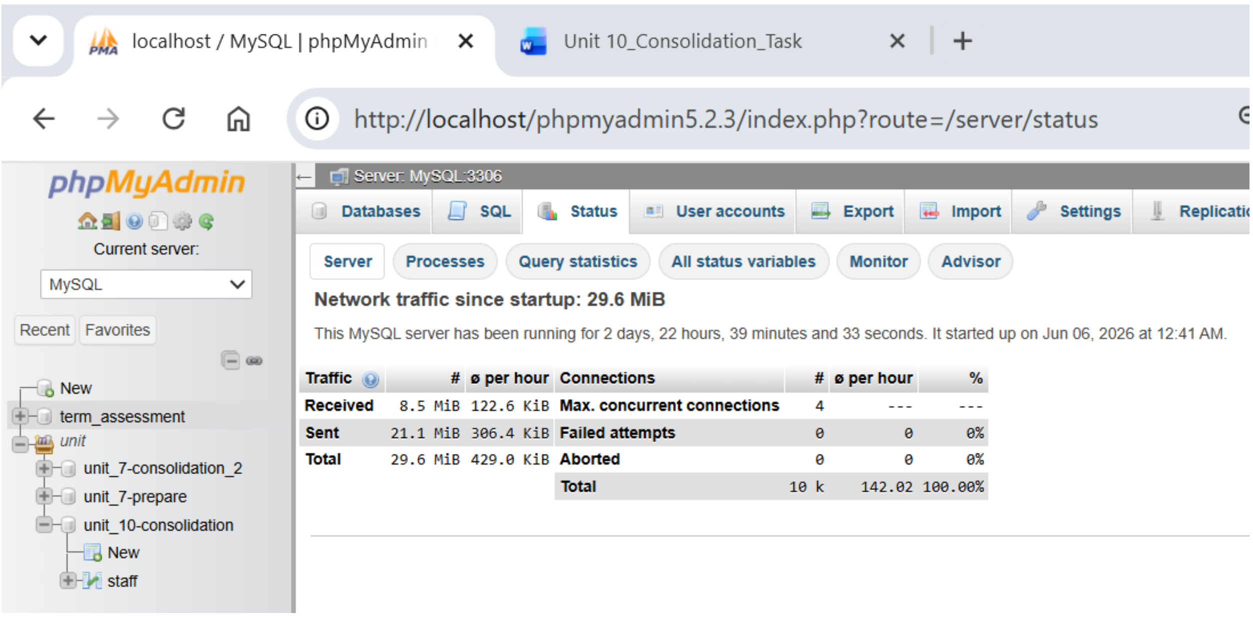
Viewing the restored data:



Task 3: Performance Monitoring

Use built-in tools (e.g., MySQL Workbench Performance Dashboard or MongoDB Atlas metrics) to monitor database performance.
Identify a scenario where the database might experience bottlenecks (e.g., high query load or large dataset).

Database performance measured via phpMyAdmin Status:



Scenario where database might experience bottlenecks

The database is a staff data management system for a medium-sized enterprise. The database might experience bottlenecks where there are multiple simultaneous connections, and too many users are logged in and trying to access data at the same time.

Bottlenecks might also occur where there are a large number of complex queries, such as payroll computing pay summaries for many individuals simultaneously.